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Metadata

Title*

Project title: Skills & Personality (SKIPPER) – Volitional Change in self-reported Social, Emotional, and Behavioral Skills

Description*

This research investigates the development of social engagement and emotional resilience skills using a quasi-experimental pretest-posttest design with two intervention groups and one control group. It is assumed that skills are more malleable than traits (Soto et al., 2021). Additionally, we investigate the role of psychological motives in skill development, i.e., achievement, power, and affiliation motives.

Participants are asked whether they would like to train one of the two skills. Based on their preference, they are randomly assigned to the preferred intervention group or the control group (conditional randomization).

We examine several research questions that may be addressed in different publications.

General research questions:
RQ1: Can social, emotional, and behavioral (SEB) skills be learned through participation in an online structured learning intervention?
RQ2: Do psychological motives play a role in the development of SEB skills (a) and in the effectiveness of the interventions (b)?
RQ3: Do SEB skills change more than the corresponding personality traits (i.e., extraversion, and emotional stability (Lechner et al., 2022; Soto et al., 2022).

The study is realized using the formr study framework (Arslan et al., 2020)

Contributors*

The authors

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Subject*

- Social and Behavioral Sciences: Psychology – Personality
- Social and Behavioral Sciences: Development Studies

Tags

- emerging adulthood (18-29 years old),
- non-cognitive skills, personality, psychological motives
- well-being

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Study Information

Hypotheses (H)*

Research questions (RQ) and hypotheses (H):

Main Research question
RQ1: Can individuals improve their emotional resilience (ER) and social engagement (SE) skills through structured learning interventions?

H1 (a) ER skills increases stronger in the ER intervention group (ER-IG) than in the control group (CG).

H1 (b) SE skills increases stronger in the SE intervention group (SE-IG) than in the CG.

We assume that emotional stability and extraversion also increase through the learning interventions, because they are the aligned personality traits of the ER and SE skills.

RQ2: Do psychological motives play a role in the development of SEB skills (a) and in the effectiveness of the interventions (b)?

H2.1: The achievement motive facilitates positive increases in ER skills.
(a) in general
(b) in the ER-IG stronger than in the CG.

H2.2: The general fear facet of motives hinders ER development.

H2.3: The affiliation motive facilitates positive increases in SE skills.
(a) in general
(b) in the SE-IG stronger than in the CG

H2.4: The power motive facilitates positive increases in SE skills.
(a) in general
(b) in the SE-IG stronger than in the CG

Fear components of the fear facets are *fear of failure*, *fear of rejection*, and *fear of losing control*.

RQ3: Are skills more malleable than traits?

H3 (a) ER skills do increase descriptively stronger than emotional stability in the ER-IG.
H3 (b) SE skills do increase descriptively stronger than emotional stability in the SE-IG.

Our cutoff for the descriptive difference is $\Delta = 0.10$.

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(RQ4: Is there a lasting effect of the intervention?)
A RQ that will be answered later.

Design Plan

Study type*

☒ Experiment - A researcher randomly assigns treatments to study subjects, this includes field or lab experiments. This is also known as **an intervention experiment** and includes randomized controlled trials. *

☐ Observational Study - Data is collected from study subjects that are not randomly assigned to a treatment. This includes surveys, “natural experiments,” and regression discontinuity designs.

☐ Meta-Analysis - A systematic review of published studies.

☐ Other

*** Please note: It is a *quasi*-experimental pretest posttest control group design with two intervention groups and one control group.**

Blinding*

Blinding describes who is aware of the experimental manipulations within a study. Mark all that apply.

☐ No blinding is involved in this study.

☒ For studies that involve human subjects, they will not know the treatment group to which they have been assigned. *[1]

☐ Personnel who interact directly with the study subjects (either human or non-human subjects) will not be aware of the assigned treatments. (Commonly known as “double blind”)

☐ Personnel who analyze the data collected from the study are not aware of the treatment applied to any given group.

***[1] Individuals can infer which groups they belong to because they are asked at the beginning whether they would like to train a skill and which one.; Based on their decision, they are randomized into the respective skill training group or into a control group.**

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Is there any additional blinding in this study?

No additional blinding.

Study design*

The study follows a quasi-experimental pretest-posttest control group design with one pre-test and two posttests. It is conducted as an online study, distinguishing between three different groups

In the pretest, participants complete a set of questionnaires. One to two days later, they receive information about personality traits as well as SEB skills. Then, they are asked whether they would like to train social engagement, emotional resilience, or no skill at all. Based on their preference, they are randomly assigned to either the corresponding intervention group (IG) (skill training) or to the control group (CG).

Over the next six weeks, participants in the IGs have the opportunity to practice their selected skill on a weekly basis, while participants in the CG are only asked to reflect on their past week and provide information about their upcoming schedules.

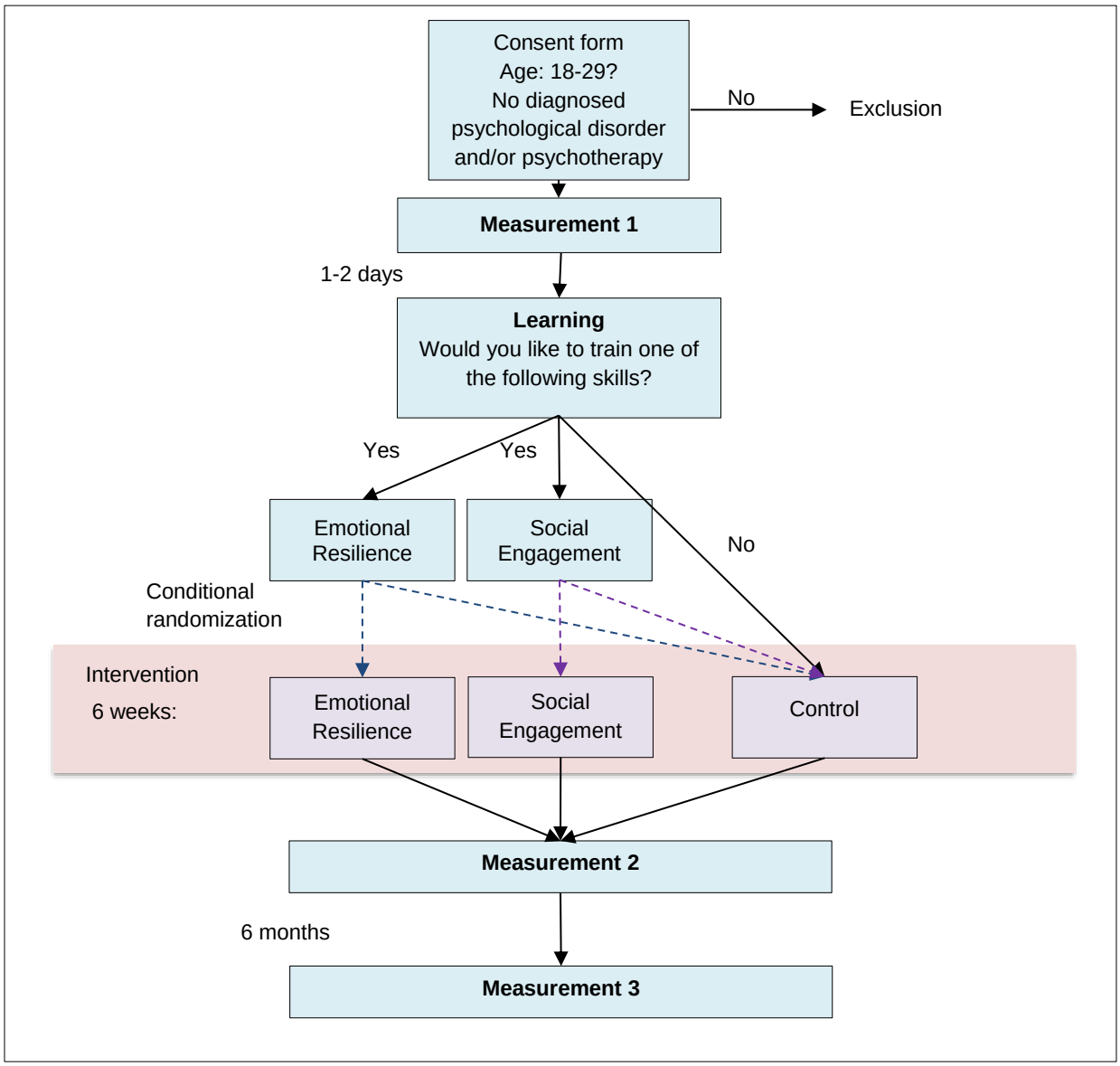
At the first posttest, participants complete questionnaires. Additionally, IG participants answer questions about the training experience. CG participants are asked whether they are interested in skill training. (They will have opportunity for a skill training later).

Six months later, a final posttest is planned as a follow-up.

➔ See Figure 1

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Figure 1: *Study design*



Randomization

Each participant is asked whether they prefer to train in emotional resilience, social engagement, or neither of these two skills. Based on their preference, they are then randomly assigned to the corresponding intervention group or the control group (conditional randomization, where self-selection plays a role).

This design reflects real-life conditions, as it considers participants' preferences rather than relying solely on controlled randomization. (external validity)

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Sampling Plan

Existing data*

☒ Registration prior to creation of data: As of the date of submission of this research plan for preregistration, the data have not yet been collected, created, or realized. *[2]

☐ Registration prior to any human observation of the data: As of the date of submission, the data exist but have not yet been quantified, constructed, observed, or reported by anyone - including individuals that are not associated with the proposed study. Examples include museum specimens that have not been measured and data that have been collected by non-human collectors and are inaccessible.

☒ Registration prior to accessing the data: As of the date of submission, the data exist, but have not been accessed by you or your collaborators. Commonly, this includes data that has been collected by another researcher or institution. *[1]

☐ Registration prior to analysis of the data: As of the date of submission, the data exist and you have accessed it, though no analysis has been conducted related to the research plan (including calculation of summary statistics). A common situation for this scenario when a large dataset exists that is used for many different studies over time, or when a data set is randomly split into a sample for exploratory analyses, and the other section of data is reserved for later confirmatory data analysis.

☐ Registration following analysis of the data: As of the date of submission, you have accessed and analyzed some of the data relevant to the research plan. This includes preliminary analysis of variables, calculation of descriptive statistics, and observation of data distributions. Please see cos.io/prereg for more information.

***[2] Data collection is ongoing, and the research project has already been preregistered (https://osf.io/sfnwt, https://osf.io/sfp9u). However, the data from the current main study has not been accessed yet.**

Explanation of existing data

// The data collection occurs simultaneously with the pre-registration of the study. //

Data collection procedures*

Participants are recruited through an email distribution list from Friedrich-Schiller University Jena (Germany), social contacts, advertisements (flyers, posters), SurveyCircle (https://www.surveycircle.com/de/), an online website (https://www.perpsy.uni-jena.de/319/skipper-

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studie), participation calls (https://www.jenatv.de/mediathek/77330/Gestalte_deine_Persoenlichkeit_aktiv_mit.html), and other methods. Data are collected online.

At the beginning of the study, participants are asked whether they are aged between 18 and 29 years. A negative response leads to an automatic exclusion from the study. Additionally, they are asked whether they have a diagnosed psychological disorder and/or currently attending psychotherapy. If so, they are also excluded before answering any questionnaires.

As psychology students at the Friedrich Schiller University Jena, they can receive two "research participation hours" (credits) if they complete the study up to the second measurement occasion. Additionally, they can participate in a voucher lottery (Wunschgutschein) if they complete the study by the last measurement occasion. All participants who complete to the study can receive individual feedback on their personality and skills profile, as well as insights into their development throughout the study.

Sample size*

Our sample size goal is 375 participants to address the main research question, determined through power analyses (Monte Carlo simulations, see below).

Sample size rationale

We conducted several a-priori power analyses to determine the required sample size. We aimed to achieve a power of 1-β = 80%, with effects of interested to be detected at a significance level of α = 0.05.

MONTE CARLO SIMULATIONS: LATENT CHANGE MODELS (LCM)

First, we used the software program R (R Core Team, 2025) to conduct Monte Carlo simulations for latent change models (LCM) to determine the required sample size for detecting stronger increases in emotional resilience (ER) and social engagement (SE) skills compared to the control group (CG).

Since the group membership is dummy-coded (i.e., group1: ER-IG = 1, CG = 0; group2: SE-IG, CG = 0), these two variables are perfectly collinear. Therefore, we conducted separate power analyses each construct (i.e., separate LCMs).

All simulations were replicated 10,000 times and run twice with different starting values (Muthén & Muthén, 2002).

Emotional resilience (ER) skills

Main research question
To detect a stronger increase of ER skills of Δ = 0.30 in the ER-IG compared to the CG, a sample size of *n* = 200 is needed (***n* = 100 per group**). We chose a difference of 0.30 because previous

studies found increases in the related emotional stability trait of approximately *d* = 0.57 (Roberts et al., 2017; Stieger et al., 2021) In our opinion, ER skills might also increase in the CG (see descriptive results of the pilot study: https://osf.io/b7q8e). Thus, we aim to investigate whether the ER intervention fosters skill increases beyond those observed in the CG.

Influence of motives:
To detect a main effect of a motive on the ER increase with a parameter size of |0.20|, a sample size of *n* = 276 would be required (n = 138 in each group). However, to detect an interaction (group x motive) with a size of |0.15|, a sample size of *n* = 420 would be needed (*n* = 210 per group).

Social engagement (SE) skills

Main research question
To detect a greater increase in SE skills of Δ = 0.20 in the SE-IG compared to the CG, *n* = 250 participants are required (**n_g = 125 per group)**. We based this effect size of 0.20 on previous studies, which reported increases in the related extraversion trait of *d* = .0.58 (Stieger et al., 2021) and *d* = 0.22 (Roberts et al., 2017) .

In contrast to the ER skills, we assume that participating in the CG will not lead to a growth in SE skills, as development SE skills requires actively seeking social interactions -something that is not explicitly encouraged in the CG.

Influence of motives:
To detect a main effect of a motive on the SE skill development with an effect size of |.20|, a sample size of *n* = 300 would be required. The sample rationale for an interaction effect (group-x-motive) of |.15| is *n* = 450.

To sum up, a total of 100 + 2*125 = 350 participants is required for the main research question. However, to ensure equally sized groups, 375 participants would be required.

GOODNESS FIT INDICES FOR LCM

Second, we estimated the sample size for the LCMs to detect specified sizes of goodness fit indices. Overall, each group requires 70 to 110 participants to achieve an adjusted goodness of fit index (AGFI) of 0.95 and a RMSEA ≤ 0.05.

POWER ANALYSES: MULTIPLE REGRESSION (ANCOVA)

Third, we used the software program G*Power as well as R to conduct a power analysis as well as R for multiple regressions and summarize the results here:

Main research question

(A) T2 ~ T1 + SELECT + GROUP

In this model, T1 and T2 are continuous variables representing the skill of interest at time 1 and time 2, respectively. The SELECT variable represents the selection of (preference for a) a skill training, and GROUP indicates group membership (categorial).

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If separate models for group comparisons (intervention vs. control group) are conducted, expecting an overall effect size of $f^2 = 0.05$ ($\approx R^2 = 0.0476$) (Cohen, 1992), a sample size of $n = 222$ would be required (i.e., 111 participants per group). Since the SEB skills show high temporal stabilities (Lechner et al., 2022) with $r = .56$ to $.80$, a greater effect size for the overall model can be expected (fixed at $R^2 = 0.50$ for a priori power analyses). For this model, fewer participants would be required.

To test whether GROUP explains variance in T2 beyond T1 and SELECT, a total of $n = 264$ participants would be required per comparison (i.e., 132 participants per group). This is based of an effect size of $f^2 = 0.03$ ($\approx \Delta R^2 = 0.0291$).

Furthermore, we conducted power analyses to detect a greater increase in the ER-IG compared to the CG of d = 0.35, and a greater increase in the SE-IG compared to the CG of d = 0.22 ($R^2 = 0.50$). Sample sizes of $n = 131$ (66 per group) and $n = 327$ (164 per group) would be required, respectively (overall: 66 + 2 * 164 = 394).

Influence of motives

(B) T2 ~ T1 + SELECT + GROUP VS. T2 ~ T1 + SELECT + GROUP + MOTIVE

To test, whether the motive variable contributes to the prediction of T2 beyond (A), with an effect of $f^2 = 0.03$ ($\approx R^2 = 0.0291$), $n = 264$ participants are required ($\frac{264}{2} * 3 = 396$)

(C) T2 ~ T1 + SELECT + GROUP + MOTIVE VS. T2 ~ T1 + SELECT + GROUP + MOTIVE + (GROUP*MOTIVE)

To test, whether the motive contributes more to an increase in a skill within an intervention group than in the control group ($f^2 = 0.03$), $n = 264$ participants are required (see above).

Stopping rule

We will undertake additional recruitment efforts to meet our sample size goal. First, we will repeat some of the procedures described above. Second, we will use other strategies, e.g. social media and social contacts. Further details will be provided.

Variables

Manipulated variables

No real manipulation is taking place. The independent variable (IV) is group membership (emotional resilience intervention, social engagement intervention, control group). It is a self-selected variable because participants choose the intervention they prefer. Based on their choice, they are the randomly assigned into their preferred group or to control group.

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Measured variables *

Below is an overview of the measured variables (Table 1). To address our research questions, we will use the following variables:

- Social, emotional, and behavioral skills (SEB skills) at time 1, time 2, and time 3.
- Achievement, affiliation, power motives, as well as the fear facet from the United Motive Scale 6
- Big Five personality traits at time 1, time 2, and time 3
- Other variables may be used, e.g. as covariates, and any such variables will be disclosed transparently at a later stage

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Table 1: *Measured variables, time points and measures*

Constructs	Time	Measure
Demographic variables	1	
Social, emotional, and behavioral skills	1,2,3	Behavioral, Emotional, and Social Skills Inventory in German language (BESSI-G, Lechner et al., 2022)
Psychological motives	1	Unified Motive Scales 6 (UMS, Schönbrodt & Gerstenberg, 2012)
Narcissism	1	Narcissistic Admiration and Rivalry Questionnaire (NARQ, Back et al., 2013)
Big Five personality traits	1,2,3	Big Five Inventory-2 (BFI-2, Danner et al., 2019)
Self-esteem	1,2,3	Three items used in the pairfam study (Huinink et al., 2011)
Self-oriented perfectionism	1	Self-oriented perfectionism (Stöber, 2012)
Subjective wellbeing	1,2,3	Adapted version of the Positive and Negative Affect Scale (PANAS, Breyer & Bluemke, 2016) Other affective states: serenity (at ease, calm, relaxed); enthusiastic (positive affect); dis-tressed (negative affect) (Röcke & Grühn, 2003)
Cognitive wellbeing	1,2,3	Life satisfaction (L1, Beierlein et al., 2015)
Preference for skill training	between 1 and 2	• Question: Would you like to train one of the following skills? ○ Emotional Resilience Skills ○ Social Engagement Skills ○ No • Question: Why (not)? • Question: How do you evaluate your skill level at the moment and which would you like to have?
Preference for another skill training	between 1 and 2	• Question: Would you like to train one of the other skills? ○ Self-Management Skills ○ Cooperation Skills ○ Innovation Skills • Question: Why? • Question: How do you evaluate your skill level at the moment and which would you like to have?
Weekly questions	6 weeks long between 1 and 2	For all groups: How was your week? What have you done? For control group: What are you going to do in the next week? For intervention groups: Please select two to three of the following exercises to train your skill next week.

Indices

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If we combine measures into indices, we will disclose this transparently.

Analysis Plan

Statistical models *

We plan to conduct latent change models (LCM) with predictors for the latent change, or use multiple regression analyses, to test our hypotheses.

LATENT CHANGE MODELS (LCMs)

For LCMs, latent change will be predicted by the group membership (GROUP: one intervention group vs. control group) and a variable SELECT, representing the self-selection to examine the RQ1.

To test whether a motive predicts change in a skill, the respective MOTIVE variable will first be used as a predictor on its own, followed by testing it with the other predictors (GROUP, SELECT), including an interaction between GROUP and MOTIVE. This will be done for each skill and group comparison separately.

To answer our RQ3 (descriptive difference in changes), we will compare the standardized change in a skill with the standardized change in the corresponding personality trait.

MULTIPLE REGRESSIONS

For the multiple regression analyses, emotional resilience skills at time 2 (T2) will be used as the dependent variable, predicted by a dummy variable GROUP specifying whether a participant is in the control group (group: CG = 0) or the respective intervention group (group: IG = 1), while controlling for emotional resilience at time 1 (T1) and possibly a variable SELECT, representing self-selection (e.g., propensity score).

Next, the MOTIVE variable will be introduced as a separate predictor in the model, first without the other predictors, and later with them. Finally, we will test for the interaction (GROUP X MOTIVE) by including this effect in the model. The same procedure will be applied to social engagement skills.

To answer the research question 3 (RQ3), we will compare descriptive changes in emotional resilience with the same changes in emotional stability descriptively within individuals in the respective intervention group (ER-IG). Similarly, we will compare changes in social engagement and extraversion within individuals in the other intervention group (SE-IG).

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Transformations

The group membership will be recoded for a better interpretation (i.e., dummy coding).

Inference criteria

We will use the standard α-level of <.05 as the criterion for determining statistical significance.

Additionally, we set a cutoff for a standardized skill increase of at least d = 0.10 within individuals in the respective intervention group. Furthermore, the standardized descriptive difference between intervention and control group should be at least 0.10 (difference between Cohens d coefficients). The difference in skill change and change in the aligned trait should also meet a cutoff criterion of 0.10.

Data exclusion

Participants who fail to answer attention check items correctly will be excluded from our analyses (e.g., “This is an attention check: Please select the answer Y.”).

Missing data

Missing data will be handled with using appropriate strategies, with full information maximum likelihood estimation (FIML) applied in latent change models.

Exploratory analysis

We will explore the data regarding descriptive statistics, including correlations, means and standard deviations. If any issues arise in our analyses, we will conduct exploratory analyses to identify potential problems.

Other

If any issues arise in our analysis, we will reduce the model complexity and/or try to test our hypotheses using alternative statistical models.

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References

Arslan, R. C., Walther, M. P., & Tata, C. S. (2020). Formr: A study framework allowing for automated feedback generation and complex longitudinal experience-sampling studies using R. *Behavior Research Methods*, *52*(1), 376–387. https://doi.org/10.3758/s13428-019-01236-y

Back, M. D., Küfner, A. C. P., Dufner, M., Gerlach, T. M., Rauthmann, J. F., & Denissen, J. J. A. (2013). Narcissistic admiration and rivalry: Disentangling the bright and dark sides of narcissism. *Journal of Personality and Social Psychology*, *105*(6), 1013–1037. https://doi.org/10.1037/a0034431

Beierlein, C., Kovaleva, A., László, Z., Kemper, C. J., & Rammstedt, B. (2015). Kurzskala zur Erfassung der Allgemeinen Lebenszufriedenheit (L-1). Advance online publication. https://doi.org/10.6102/zis229

Breyer, B., & Bluemke, M. (2016). Deutsche Version der Positive and Negative Affect Schedule PANAS (GESIS Panel). Advance online publication. https://doi.org/10.6102/zis242

Cohen, J. (1992). A power primer. *Psychological Bulletin*, *112*(1), 155–159. https://doi.org/10.1037/14805-018

Danner, D., Rammstedt, B., Bluemke, M., Lechner, C., Berres, S., Knopf, T., Soto, C. J., & John, O. P. (2019). Das Big Five Inventar 2. *Diagnostica*, *65*(3), 121–132. https://doi.org/10.1026/0012-1924/a000218

Huinink, J., Brüderl, J., Nauck, B., Walper, S., Castiglioni, L., & Feldhaus, M. (2011). Panel analysis of intimate relationships and family dynamics (pairfam): Conceptual framework and design. *Zeitschrift Für Familienforschung*, *23*(1), 77–101. https://www.ssoar.info/ssoar/handle/document/37646

Lechner, C. M., Knopf, T., Napolitano, C. M., Rammstedt, B., Roberts, B. W., & Soto, C. J. (2022). The behavioral, emotional, and social skills inventory (BESSI): Psychometric properties of a German-language adaptation, temporal stabilities of the skills, and associations with personality and intelligence. *Journal of Intelligence*, *10*(3). https://doi.org/10.3390/jintelligence10030063

Roberts, B. W., Luo, J., Briley, D. A., Chow, P. I., Su, R., & Hill, P. L. (2017). A systematic review of personality trait change through intervention. *Psychological Bulletin*, *143*(2), 117–141. https://doi.org/10.1037/bul0000088

Röcke, C., & Grühn, D. (2003). *German translation of the PANAS-X*. Unpublished manuscript, Free University Berlin, Berlin, Germany.

Schönbrodt, F. D., & Gerstenberg, F. X. (2012). An IRT analysis of motive questionnaires: The Unified Motive Scales. *Journal of Research in Personality*, *46*(6), 725–742. https://doi.org/10.1016/j.jrp.2012.08.010

Soto, C. J., Napolitano, C. M., & Roberts, B. W. (2021). Taking skills seriously: Toward an integrative model and agenda for social, emotional, and behavioral skills. *Current Directions in Psychological Science*, *30*(1), 26–33. https://doi.org/10.1177/0963721420978613

Soto, C. J., Napolitano, C. M., Sewell, M. N., Yoon, H. J., & Roberts, B. W. (2022). An integrative framework for conceptualizing and assessing social, emotional, and

behavioral skills: The BESSI. *Journal of Personality and Social Psychology*, *123*(1), 192–222. https://doi.org/10.1037/pspp0000401

Stieger, M., Flückiger, C., Rüegger, D., Kowatsch, T., Roberts, B. W., & Allemand, M. (2021). Changing personality traits with the help of a digital personality change intervention. *Proceedings of the National Academy of Sciences of the United States of America*, *118*(8). https://doi.org/10.1073/pnas.2017548118

Stöber, J. (2012). Selbstbezogener Perfektionismus - Schüler [Fragebogenskala: Version 1.0]. Advance online publication. https://doi.org/10.7477/10:20:1